

HSTS Project Run Guide

Windows PowerShell • G7_Server.jar and G7_Client.jar • Java 17+

Start order MySQL → Server → Client(s)	Default port 5555	Client platform Windows
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Prerequisites

- 1** Install Java
Use JDK 17 or newer. Verify with `java -version`.
- 2** Install MySQL
Use MySQL 8 and create the `hsts_prototype` database.
- 3** Open the project
Run commands from `C:\courses\SoftwareEngineering\HSTS-Prototype` or your cloned repository folder.

Build the runnable JARs

Stop any running HSTS server/client before a clean build, then run:

```
cd C:\courses\SoftwareEngineering\HSTS-Prototype
mvn clean package
```

Expected output: `target\G7_Server.jar` and `target\G7_Client.jar`.

Start the server

Open PowerShell in the project folder. Replace placeholders locally; never commit passwords or API keys.

```
$env:HSTS_DB_URL = "jdbc:mysql://localhost:3306/hsts_prototype?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC"
$env:HSTS_DB_USER = "root"
$env:HSTS_DB_PASSWORD = "<YOUR_MYSQL_PASSWORD>"
$env:HSTS_SERVER_PORT = "5555"
$env:HSTS_BOT_PROVIDER = "DETERMINISTIC"

java -jar target\G7_Server.jar
```

Keep the server terminal open. Startup initializes or migrates the database safely.

Start a client

Open a second PowerShell terminal:

```
cd C:\courses\SoftwareEngineering\HSTS-Prototype
$env:HSTS_SERVER_HOST = "localhost"
$env:HSTS_SERVER_PORT = "5555"

java -jar target\G7_Client.jar
```

Start additional clients in separate terminals using the same commands. Different users may connect simultaneously; duplicate login for the same account is rejected.

Optional: Gemini Course Bot

Only the server operator needs the Gemini key. Client users never receive it.

```
$env:HSTS_BOT_PROVIDER = "GEMINI"
$env:HSTS_GEMINI_API_KEY = "<YOUR_GEMINI_API_KEY>"
$env:HSTS_GEMINI_MODEL = "gemini-3.5-flash-lite"
$env:HSTS_GEMINI_TIMEOUT_SECONDS = "30"

java -jar target\G7_Server.jar
```

For an offline demonstration, use HSTS_BOT_PROVIDER=DETERMINISTIC and no API key.

Connecting from another computer

Run the server on one machine. On every client machine, replace localhost with the server computer’s LAN IP and allow TCP port 5555 through the server firewall.

```
$env:HSTS_SERVER_HOST = "<SERVER_LAN_IP>"
$env:HSTS_SERVER_PORT = "5555"
java -jar G7_Client.jar
```

Common startup problems

Message / symptom	What to check
Access denied for user / using password: NO	Set HSTS_DB_USER and HSTS_DB_PASSWORD in the same terminal before starting the server.
Connection refused	Start the server first; confirm host, port, firewall, and that the server terminal remains open.
JAR appears outdated	Close HSTS processes and run mvn clean package again.
JavaFX native-access warning	This is currently non-fatal; the client should still start.
Gemini unavailable	Check the server’s model/key configuration, or switch to DETERMINISTIC mode.

Security: Never place database passwords or Gemini API keys in source code, README examples, screenshots, commits, or client configuration. Each server operator supplies secrets through environment variables.